NUI Galway
OÉ Gaillimh

Semester 1 Examinations 2017 / 2018

Exam Code(s)	4BCT1, 4BSE1, 1MSME1,1MEEE1,1MECS1, 1CSD1, 1CSD2, 2SPE1
Exam(s)	B.Sc. Degree (Computer Science and Information Technology) B.E. Energy Systems Engineering MSc in Mechanical Engineering ME (Electrical and Computer Engineering) ME (Electrical and Electronic Engineering) M.Sc. in Computer Science (Data Analytics)
Module Code(s)	CT561
Module(s)	Systems Modelling and Simulation
Paper No.	I
External Examiner(s)	Dr. Jacob Howe
Internal Examiner(s)	Dr. Michael Schukat *Dr. Jim Duggan

Instructions:

Answer any 3 questions. All questions carry equal marks.

Duration	2hrs
No. of Pages	5 (Including Cover Page)
Department(s)	Information Technology
Requirements	Graph Paper

1. (a) List the three general principles of stock and flow systems, and illustrate these using a one-stock model of **Customers**, with appropriate flows.

[5]

(b) Construct a stock and flow model from the following description of a University. Specify the stock equations (there is no need to specify flows or auxiliaries). Clearly show and label feedback loops in the model.

- The university has **students**, and these are added to by **enrolments**, and depleted via **attrition** and **graduations**.
- The University has **staff**, and these are added to by **hires**, and depleted by **retirements** and **quits**.
- An important indicator is the **student to staff ratio**.
- An increase in **student to staff ratio** results in:
    - A reduction in **enrolments**
    - An increase in **attrition**
    - An increase in **hires**
    - An increase in **quits**

[8]

(c) Based on the stock and flow model in part(b):

- Show the main feedback loops (that involve only one stock) and calculate their polarity, by tracing an increase in a variable all the way around a feedback loop, and observing its direction of change.
- Discuss the impact each of these loops could have on the recruitment of students into the University. What steps should management take in order to ensure that the University can thrive with a scenario of increasing student numbers?

[12]

2. (a) Consider the derivative d$S/dt = t$

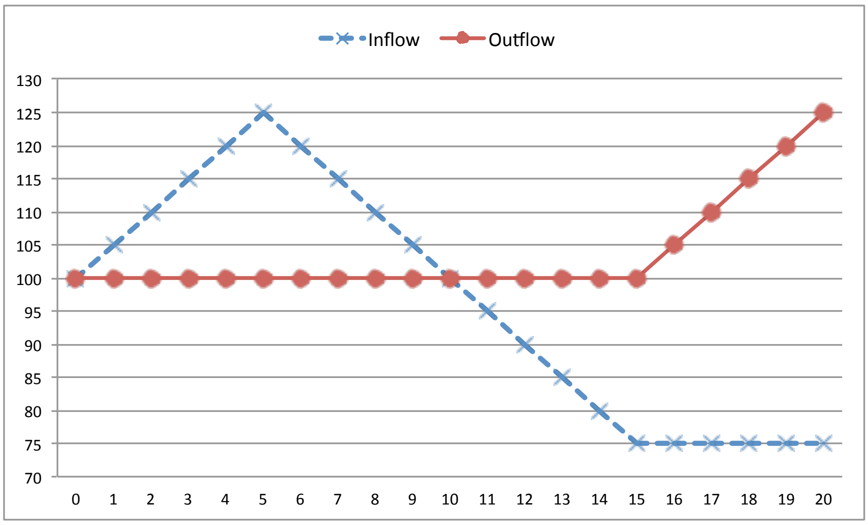
Show this as a stock and flow model, with equations, assuming the initial value of the stock is 0. Show what (shadow) variable in Vensim can be used to represent the time (t).

Integrate the derivative analytically, and use this result to calculate the value of the stock after 20 time units.

[8]

(b) Based on the flow variables shown in the diagram, and assuming that the initial value of the stock is 150, perform the following:

- Calculate, and show, the net flow
- Use graphical integration to plot the stock over time, clearly showing the behaviour mode for each segment (e.g. *increasing at an increasing rate, etc.)*



[12]

(c) Show where the inflection points are on the Stock trajectory. Discuss why an analysis of infection points can be useful in system dynamics models.

[5]

3. (a) Show clearly how the following expression can be derived from an SIR model in order to calculate whether or not a disease outbreak will occur in a totally susceptible population (i.e. where $S = N$, C_E is the effective contract rate, and D is the recovery delay).

$$C_E D > 1$$

[7]

(b) Design a stock and flow model (with equations) to simulate the spread of influenza. The model should have the following features:

- Its core structure should be based on the *Susceptible – Infected – Recovered* model.

- The effective contact rate is 2.

- There are two types of Infection, (1) Mild and (2) Severe. Individuals with a mild infection recover after 2 days, whereas those with a severe infection will take 10 days to recover. Both infection types are equally contagious.

- A limited resource known as *antivirals* can be administered to help severely infected people, and this will lead to a change to a mildly infected state. When antivirals are administered, the number of available antivirals is reduced.

- An auxiliary called *fraction severe* is used in the model to determine how many people are severely infected. This value ranges between [0,1].

[12]

(c) Show how the stock and flow model would need to be modified in order to model vaccinations as a consumable resource, where the vaccine efficacy is also included in the model (i.e. the percentage of times on average the vaccine is successful).

[6]

4. (a) Describe Little's Law, and why it is useful in helping understand the behaviour of systems in equilibrium.

[3]

(b) If a broadband provider has a pending pool of 5,000 installations to be performed, and it installs an average of 2,500 per month, what is the average waiting time (assume equilibrium).

Draw a stock and flow model to illustrate your answer.

[3]

(c) In the context of managing a stock, describe the difference between the *anchor* and the *adjustment*.

[4]

(d) Create a stock and flow model (with all equations) of staff recruitment for a software company, where the hire rate is based on the stock management structure. Assume there are three types of employee (Rookie, Senior, Consultant – an aging chain structure), and that attrition can happen at each stage.

- The target number of employees is 300
- There are 100 employees in each category
- The progression delay is 24 months (Rookie-Senior) and 48 months (Senior-Consultant).
- The attrition rate is 5% for Rookies, 3% for Senior, and 1% for Consultants.

[12]

(e) How could the equilibrium values for the model in part (d) be estimated using simulation?

[3]